GIS-Based Farm Viability Assessment

Smallholder Farmer Onboarding for Kachia LGA, Kaduna State, Nigeria.

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1. Introduction

ThriveAgric is a fast-growing agricultural technology company dedicated to ensuring food security while building the largest network of profitable smallholder farmers across Africa. One of ThriveAgric's operational model is Farm Mapping; the systematic identification, measurement, and viability assessment of farmlands, this is done prior to input financing and loan disbursement. This process ensures that agricultural loans are disbursed toward farms with the highest potential for yield and repayment.

This project presents a GIS-based Farm Viability Assessment for Kachia Local Government Area (LGA) in Kaduna State. A region with significant farming activity and one of ThriveAgric's key operational zones. Using spatial analysis techniques in QGIS, 24 sample farms were assessed across four environmental suitability factors (soil texture, slope, proximity to water, and land cover) to generate a Farm Viability Score for each farm, directly supporting ThriveAgric's loan disbursement decision-making process.

2. Objectives

- Assess the agricultural viability of 24 sample 'smallholder' farms in Kachia LGA using GIS-based spatial analysis
- Generate a composite Farm Viability Score for each farm based on slope, water proximity, soil texture, and land cover suitability
- Classify 'smallholder' farms into viability categories to support ThriveAgric's loan disbursement and farm onboarding decisions
- Produce professional cartographic outputs suitable for field officer briefings and stakeholder reporting

3. Study Area

Kachia LGA is located in the southern part of Kaduna State, Nigeria. The LGA is predominantly rural with smallholder farming as the primary livelihood activity. It is renowned as a global hub for ginger production. Other crops grown in the area include maize, sorghum, groundnuts, and these are all crops supported under ThriveAgric's input financing programme.

The area is characterised by gently undulating terrain with some steep ridges in the central zone, a network of rivers and seasonal waterbodies, and soils dominated by Sandy Loam and Clay Loam textures. Land cover is predominantly cropland and grassland, indicating significant existing agricultural activity across the LGA.

4. Data Sources

Dataset	Source	Description	Resolution
Digital Elevation Model	USGS EarthExplorer	SRTM 1 Arc-Second DEM used to derive slope	30m
Land Cover	ESA WorldCover 2021	10-class global land cover classification	10m
Soil Texture	ISRIC SoilGrids (AfSIS)	USDA soil texture classes for Africa	250m
River Network	OpenStreetMap (OSM)	Polyline river features for Kachia LGA	Vector
Waterbodies	OpenStreetMap (OSM)	Polygon waterbody and dam features	Vector
Farm Polygons	Google Earth Pro (Digitized)	24 sample farms simulating onboarding data	Vector

5. Methodology

All spatial data was processed and analysed in QGIS 3.40 (Bratislava). All datasets were reprojected to WGS 1984 UTM Zone 32N (EPSG:32632) to ensure accurate distance and area calculations in hectares. The analysis followed a six-phase workflow:

5.1 Data Preparation

- All datasets clipped to Kachia LGA administrative boundary
- Vector layers reprojected to EPSG:32632 using standard projection tools
- Raster layers reprojected using Warp (Reproject) and Nearest Neighbour resampling for categorical data, Bilinear for continuous data
- 24 sample ‘smallholder’ farm polygons digitized in Google Earth Pro over visible cropland, imported as KML and converted to shapefile

5.2 Suitability Factor Analysis

Four environmental factors were selected based on ThriveAgric’s farm mapping criteria. Each factor was reclassified into a 1–5 suitability score:

Factor	Source	Scoring Criteria	Weight
Soil Texture	ISRIC SoilGrids	Loam = 5 (Best); Sandy Clay = 1 (Poor)	35%
Slope	USGS SRTM	0–5° = 5 (Best); >20° = 1 (Poor)	25%
Distance to Water	OSM Rivers & Dams	0–500m = 5 (Best); >5,000m = 1 (Poor)	25%

Land Cover	ESA WorldCover	Cropland = 5 (Best); Urban/Water = 1 (Poor)	15%

Soil was weighted highest (35%) as the primary determinant of crop yield. Slope and water proximity were equally weighted (25% each) given their combined importance for mechanisation and irrigation. Land cover received the lowest weight (15%) as it reflects current state rather than permanent agricultural potential.

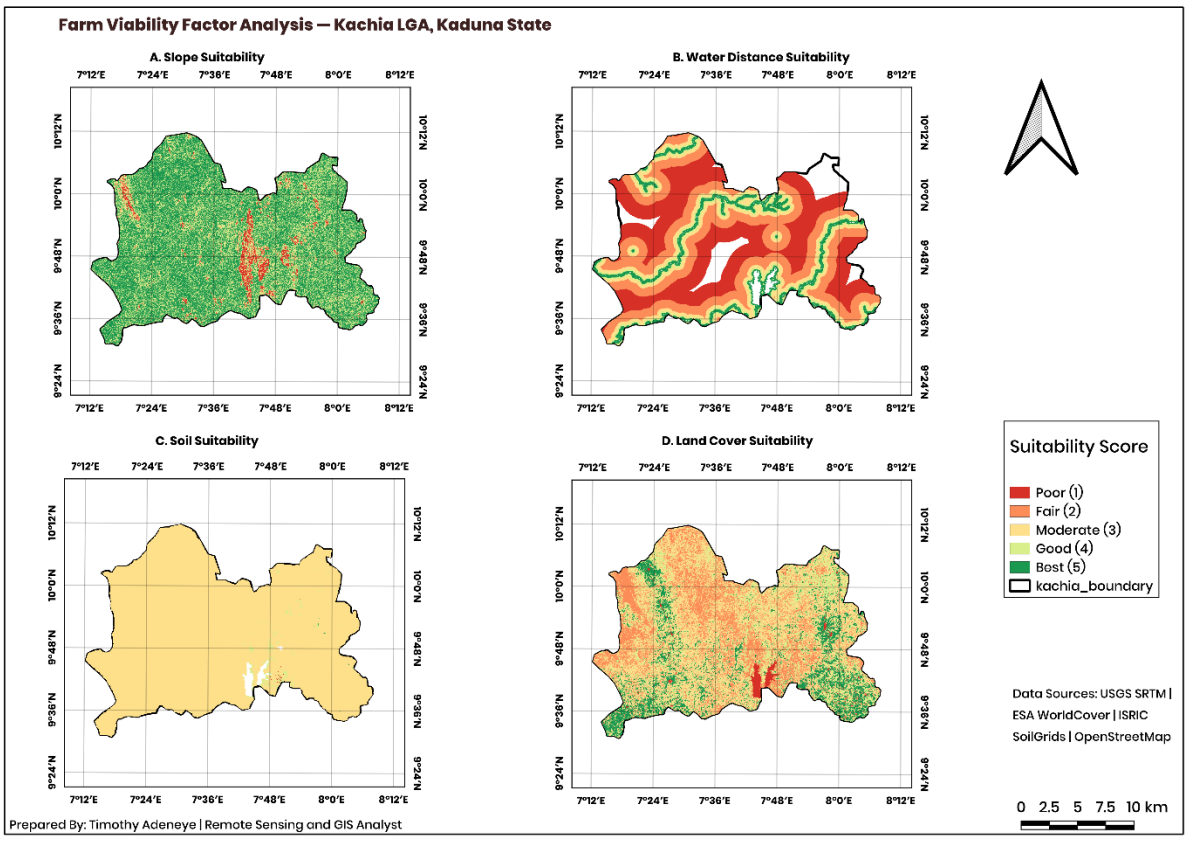

Figure 1: Farm Viability Factor Analysis - Kachia LGA, Kaduna State Map

5.3 Proximity Analysis

River and waterbody layers were rasterized separately at 30m resolution. The GDAL Proximity tool was applied to generate Euclidean distance surfaces for each. Both distance rasters were combined using the Raster Calculator minimum () function to produce a single water proximity raster representing distance to the nearest water source.

5.4 Weighted Overlay

The four reclassified suitability rasters were combined in the QGIS Raster Calculator using the following formula:

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Farm Viability Score = (Soil × 0.35) + (Slope × 0.25) + (Water ×
0.25) + (Land Cover × 0.15)
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The resulting raster represents a continuous Farm Viability Score ranging from 1 (lowest viability) to 5 (highest viability) across all of Kachia LGA.

5.5 Farm Scoring and Classification

Zonal Statistics extracted the mean viability score for each of the 24 sample farm polygons from the composite viability raster. Individual factor scores were also extracted per farm. Farms were classified into viability categories based on their Final Score:

Viability Class	Score Range	Loan Recommendation
High	≥ 4.0	Full loan disbursement recommended
Medium	3.0 – 3.99	Loan disbursement with standard conditions
Low	< 3.0	High risk; reduced loan or field verification required

6. Results and Findings

6.1 Farm Viability Distribution

Farm Viability Scores across the 24 sampled farms ranged from 2.64 to 3.65, with a mean score of approximately 3.10. No farms achieved High viability, reflecting the significant environmental constraints — particularly water proximity — across Kachia LGA.

Viability Class	Number of Farms	Percentage	Implication
High	0	0%	No farms met optimal conditions across all factors
Medium	16	67%	Suitable for loan disbursement with monitoring
Low	8	33%	Elevated risk; requires field verification

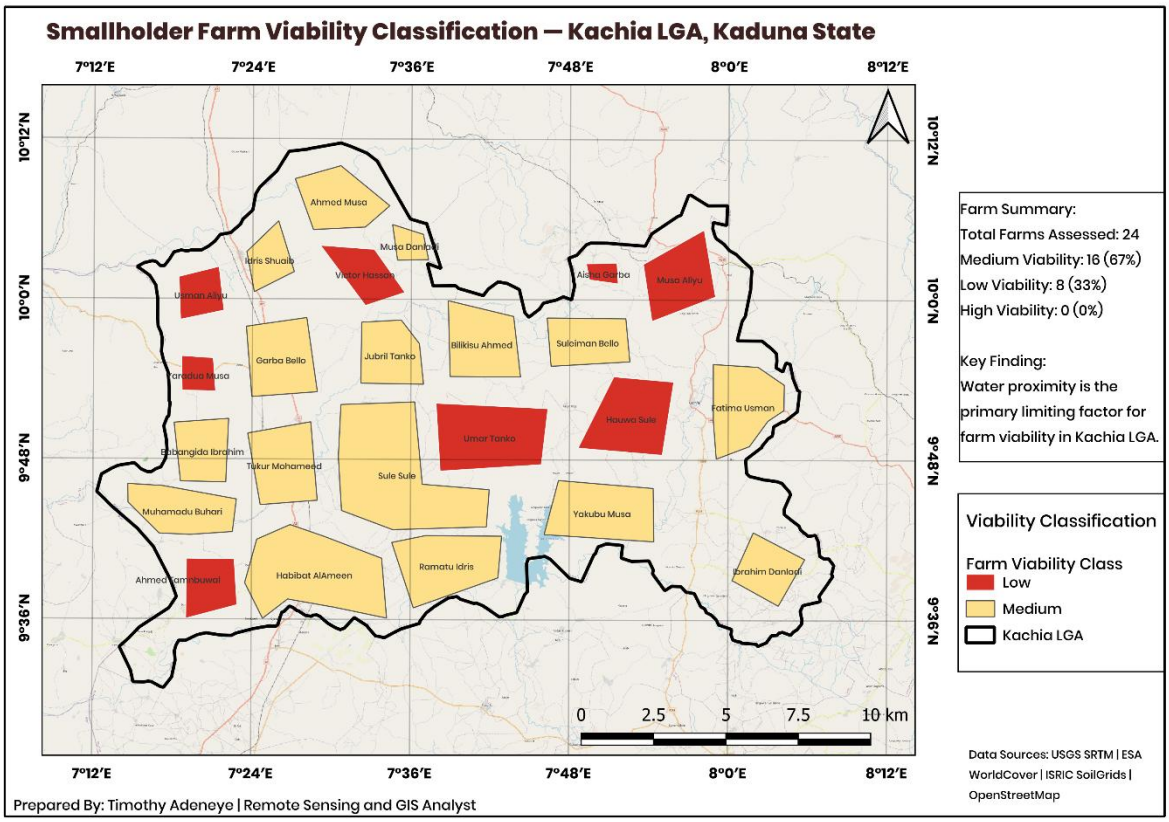

Figure 2: Smallholder Farm Viability Classification - Kachia LGA, Kaduna State Map

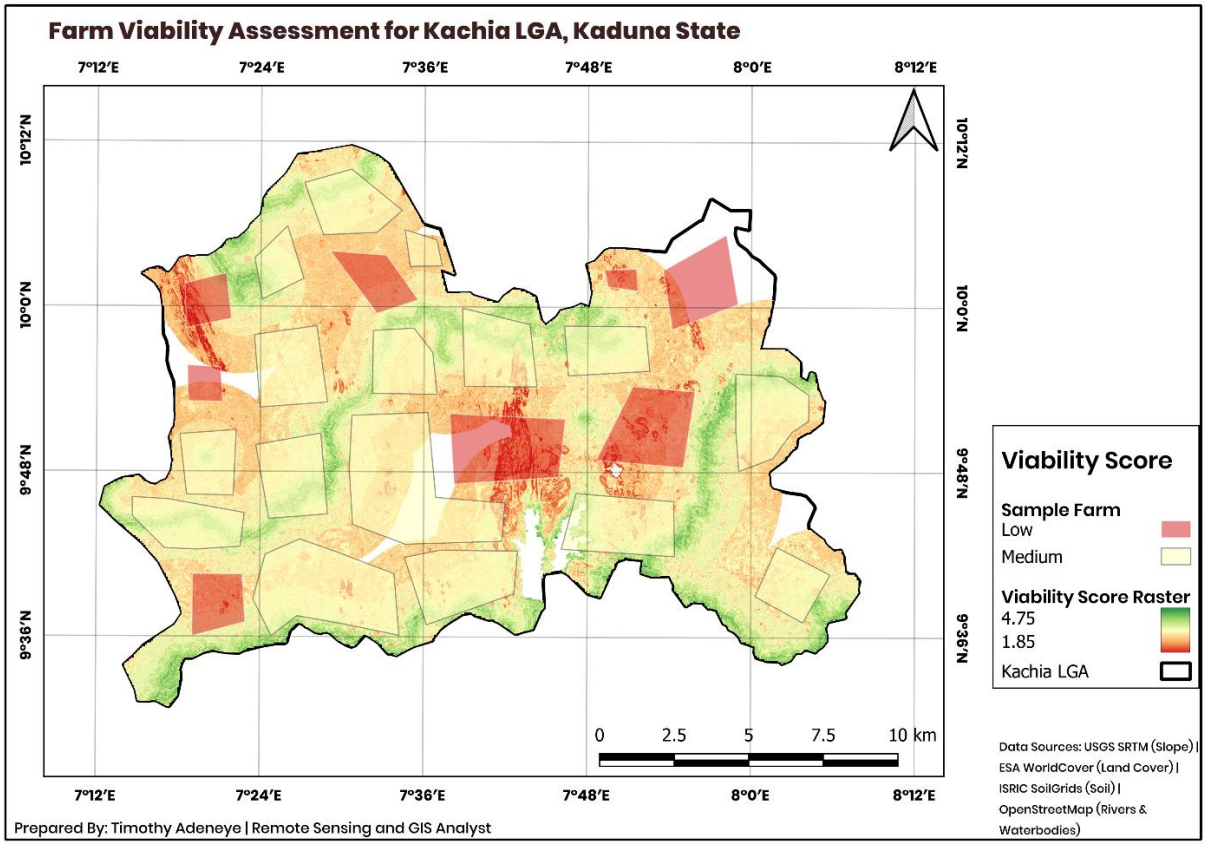

Figure 3: Farm Viability Assessment for Kachia LGA, Kaduna State Map

6.2 Key Findings

- Water proximity is the primary limiting factor. Most of Kachia LGA lies more than 5km from any river or waterbody, resulting in low water suitability scores (1–2) for most farms
- Slope is not a significant constraint. Kachia is predominantly flat to gently rolling terrain, with most farms scoring 4–5 for slope suitability, making the area well-suited for mechanised farming
- Soil texture is uniformly moderate. Sandy Loam and Clay Loam dominate across the LGA, scoring 3–4, indicating adequate but input-dependent agricultural potential
- Land cover is favourable. The predominance of cropland and grassland results in high land cover suitability scores (4–5), confirming active farming across the study area
- 67% of sampled farms are viable for ThriveAgric financing under standard conditions, representing a significant opportunity for loan disbursement and farmer onboarding in Kachia LGA

7. Recommendations

- Prioritise Medium viability farms located within 1–2km of rivers or waterbodies for immediate loan disbursement — these represent the highest return-on-investment opportunity.
- Implement targeted irrigation support for Low viability farms in water-scarce zones — improving water access could elevate these farms to medium viability.
- Focus input financing on farms with Clay Loam soils which offer the best balance of moisture retention and fertility for maize and sorghum cultivation.
- Conduct field verification for all 8 Low viability farms before disbursing loans. On-ground assessment may reveal localised conditions not captured in satellite derived data.
- Expand this viability model to additional LGAs in Kaduna State and other locations to support ThriveAgric's regional onboarding and expansion strategy.

8. Tools and Technologies

Tool	Purpose
QGIS 3.40 (Bratislava)	Primary GIS platform — data processing, spatial analysis, map production
GDAL Proximity Tool	Euclidean distance raster generation for river and waterbody layers
GDAL Rasterize	Vector to raster conversion for proximity analysis inputs
QGIS Raster Calculator	Weighted overlay and raster algebra operations
QGIS Zonal Statistics	Extraction of mean viability scores per farm polygon
Google Earth Pro	Digitizing of 24 sample farm polygons across Kachia LGA
QGIS Print Layout	Professional cartographic output production (3 maps)

9. Limitations

- Farm polygons are simulated from Google Earth Pro imagery and do not represent live farms. They are hypothetical just for the purpose of the analysis and results should be validated with surveyed farm boundaries.
- ISRIC SoilGrids data is at 250m resolution and may not capture local soil variability at the individual farm level.
- The analysis does not account for seasonal variability in water availability. Rivers shown as perennial on OSM may be seasonal in practice.
- Rainfall and temperature data were not incorporated. Future iterations should include climatic suitability factors for a more comprehensive assessment.

10. Conclusion

This GIS-based Farm Viability Assessment demonstrates how spatial analysis can directly support ThriveAgric's farm onboarding and loan disbursement workflow. By combining terrain, soil, water proximity, and land cover data into a single composite viability score, field officers and analysts can make faster, more evidence-based decisions about which farms to prioritise for financing.

The finding that 67% of sampled farms in Kachia LGA meet the threshold for loan disbursement under standard conditions represents a significant opportunity for ThriveAgric's expansion in Kaduna State. The identification of water proximity as the primary limiting factor provides a clear, actionable basis for targeted irrigation investment to unlock the remaining 33% of Low viability farms.

This methodology is scalable and replicable across several states in Nigeria and beyond, providing a robust and cost-effective framework for evidence-based agricultural financing at scale.